GOOGLE CLOUD AI BUILDER CUP 2026 · RETAIL & COMMERCE

# Taal

*The agent that sells what the forecast says you'll throw away.*

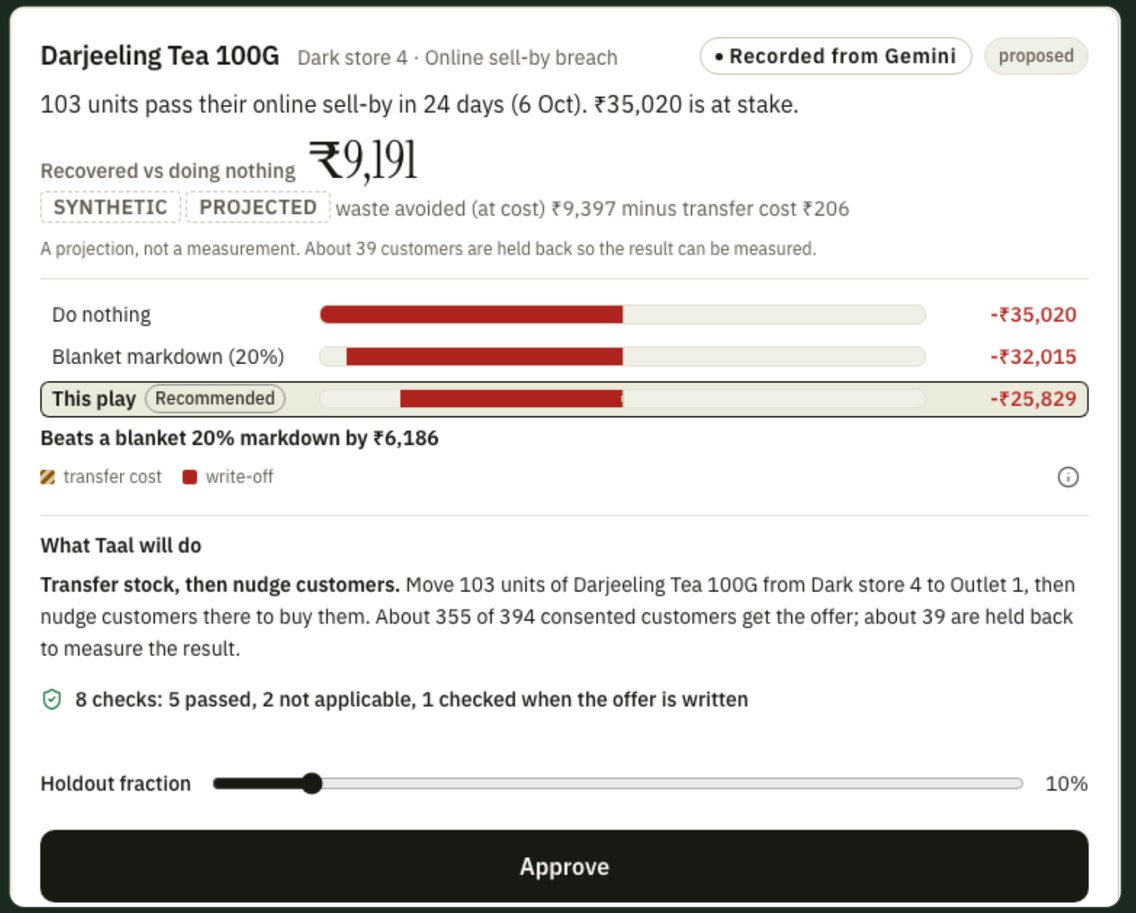
**Taal spots stock at risk, plans a play with Gemini, a person approves, the offer goes out, and a holdout measures it. The agent is refused by code until its play carries a holdout; code assigns the groups, and the app says plainly when the result has not proved anything yet.**

Context: Blinkit-scale inventory loss is about ₹0.5 cr per store a year (all causes). Taal projects ₹2.83 lakh per node a year at a 5% response rate (synthetic tenant).

**₹9,191** projected recovery vs doing nothing on one tea lot; +₹6,186 vs a blanket markdown PROJECTED

For a dark-store manager clearing 103 packs of Darjeeling tea, 24 days from sell-by, ₹35,020 at stake. Synthetic data, labelled.

**Nandish** · lead **Rinu** · co-builder




The live Play Desk, 16 Oct: the tea play, priced by code, waiting for a person to approve

THE PROBLEM · HOW BIG, WHICH RULE, WHO DECIDES

# The forecast sees the write-off coming, but nothing acts on it.

HOW BIG · AS REPORTED

**1.8%**

**of net order value (NOV) is lost to inventory at Blinkit.**

As reported by Inc42, 22 Jul 2026. Per-store line below shows its inputs, from the same article.

**₹17,132 cr ÷ 2,443 stores × 4 × 1.8% ≈ ₹0.5 cr per store a year**

All inventory loss, one quarter annualised. Context for scale, not a Taal result.

THE RULE · AN ADVISORY, NOT LAW

**30% or 45 days**

**of shelf life must remain at delivery of an online order.**

FSSAI advisory to e-commerce food sellers, 3 Dec 2024. Stock can be in date on the shelf yet past its online sell-by, so the deadline can come weeks before expiry.

Taal plans to that online sell-by line, not the printed expiry. Either-limit reading is its configurable default.

WHO IT IS FOR

## A dark-store manager with a sell-by lot

103 packs of Darjeeling tea, 24 days before the online sell-by, ₹35,020 at stake (seeded tenant).

**A demand planner** checks the guardrails and approves the play.

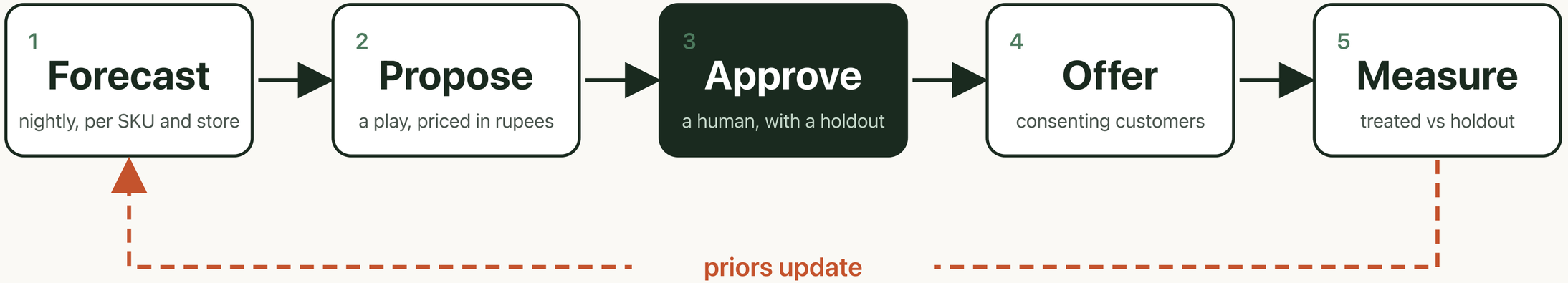
**The category or supply-chain head** owns write-off and buys Taal.

NEXT The loop and the hero result are on slide 3; projected impact is on slide 15; 3 of 7 practitioners we asked write off weekly or daily (slide 13).

THE LOOP AND THE HERO PLAY

# One worked example: a transfer plus targeted nudge recovers ₹9,191 vs doing nothing on one tea lot

PROJECTED · SEEDED TENANT




TEA · GEMINI RUN · PROJECTED

**₹9,191**

transfer plus targeted nudge (₹9,397 waste avoided – ₹206 transfer); ₹6,186 ahead of a blanket 20% markdown

HOLDOUT · SYNTHETIC ORDERS

**+0.3 pp**

INCONCLUSIVE, SYNTHETIC (pp = percentage points; 95% CI −8.7 to +1.6). A 5 pp lift needs about 1,156 customers (SIMULATION).

WHAT THIS PROVES

**The loop closes**

Measure runs end to end on synthetic orders; effect size needs a real pilot

THE HEADLINE

The first value is the right decision (transfer vs markdown); the holdout proves the offer part. Why tea: a generic dark-store tenant, so no employer data is used.

WHAT IS NEW · DIFFERENTIATION

# No play without a holdout, enforced in code.

### WHAT CLEARANCE TOOLS DO (PUBLIC VENDOR PAGES, READ 9 OCT)

- Price near-expiry stock down from its expiry date, in store
- Optimise markdowns and promotions inside enterprise planning suites
- Sell surplus food to consumers through a marketplace

### WHAT TAAL ADDS

- Plans to the online sell-by line, not the printed expiry
- Every play carries a code-enforced holdout; an inconclusive result is reported as inconclusive and the measured rupee figure is withheld
- The approved play is written back into the forecast; each safety layer is measured separately

**1 · Gemini proposes**

mechanic, audience and rationale for the at-risk lot, in an ADK loop

**2 · Code refuses it without a holdout**

The rule (rule_holdout_required): reject if the holdout fraction is outside [0.05, 0.9], min_treated_n is unset, or the seed is missing; code then sets the seed and assigns the groups

**3 · The result says what it proved**

Tea: +0.3 pp, 95% CI −8.7 to +1.6, INCONCLUSIVE, SYNTHETIC. About 1,156 customers needed for a 5 pp lift (SIMULATION)

**Each safety layer measured on its own:** the tool refused every cross-customer call (the model tried 3 of 3 times); Model Armor blocked 3 of 8 chat injection cases and fails open; the cropped-label case (V6) closed in code on 16 Oct. **Method:** protocol committed before any model call, a run is INVALID if a fallback model answers, spend cap enforced. **Why it compounds:** every measured play updates that retailer's response prior for its mechanic and category; a tool that skips the holdout has nothing to learn from.

LIST OF FEATURES

# Six capabilities, gated by rules a model never touches.

### Vision intake

Photograph a pallet; dates and counts come back with confidence. Low confidence goes to a person.

### Nightly Sense

Per-SKU, per-node forecasts. Six gap types, each with rupees and a deadline.

### Planner Agent

ADK LoopAgent, max 3 iterations, six deterministic tools; a Cost Governor picks the gaps.

### Approve + assignment

A human approves. Holdout by hash; offers go to treated customers only.

### Customer Agent

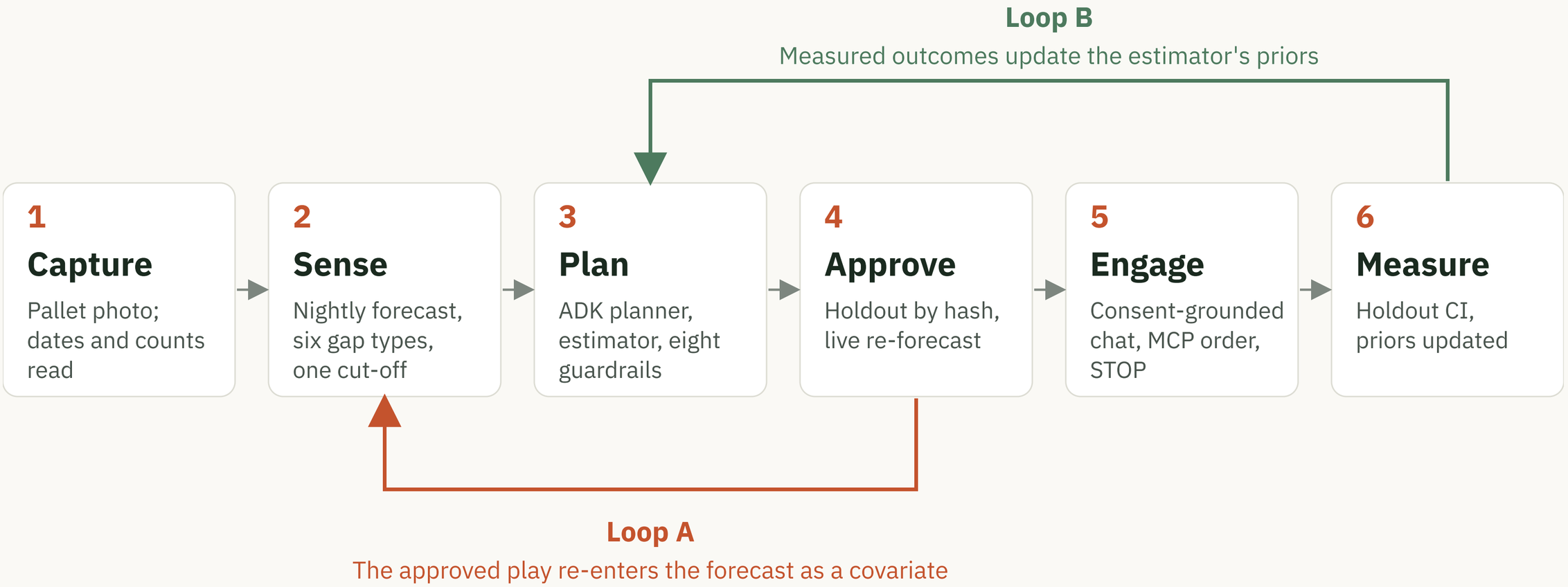
Grounded in stock, offers and consent. Orders through an in-process MCP tool.

### Measure

Treated vs holdout, 95% CI. Priors update from the counts.

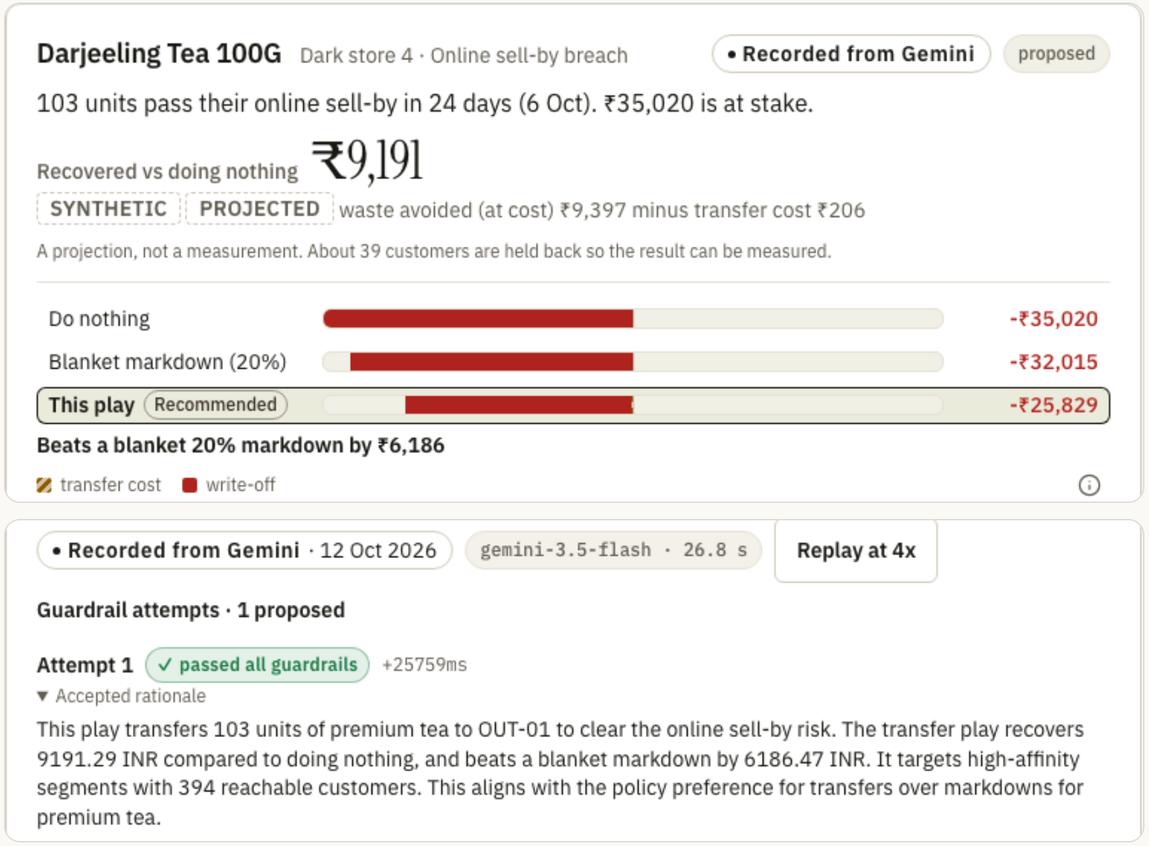
PROCESS FLOW

# Two closed loops, not an open-ended pipeline.



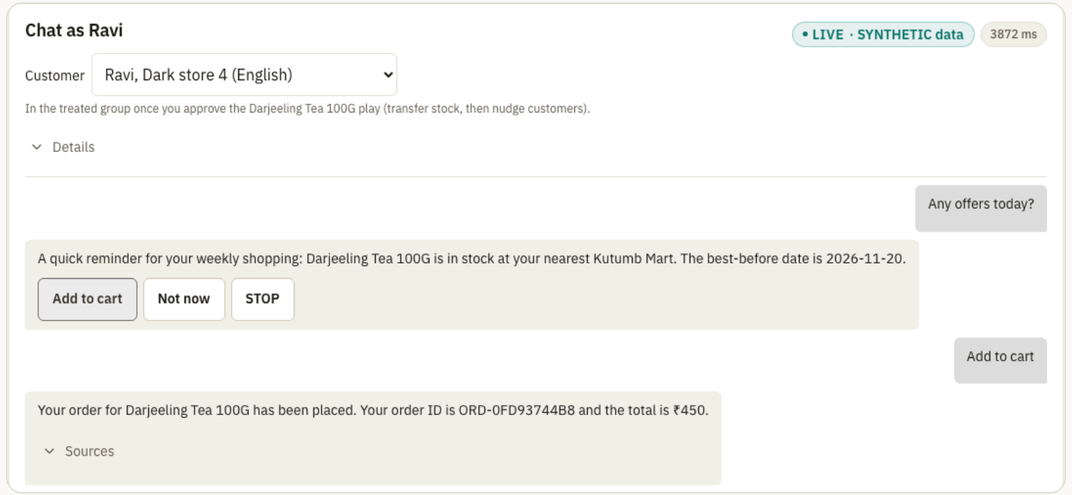
SCREENS

# Gemini drafts the play; code prices every rupee; the customer orders in chat.




## Play Desk, live app, 16 Oct

Gemini drafted the play and its rationale (recorded 12 Oct); code priced every rupee and the guardrails passed it on the first attempt. Projected, synthetic.

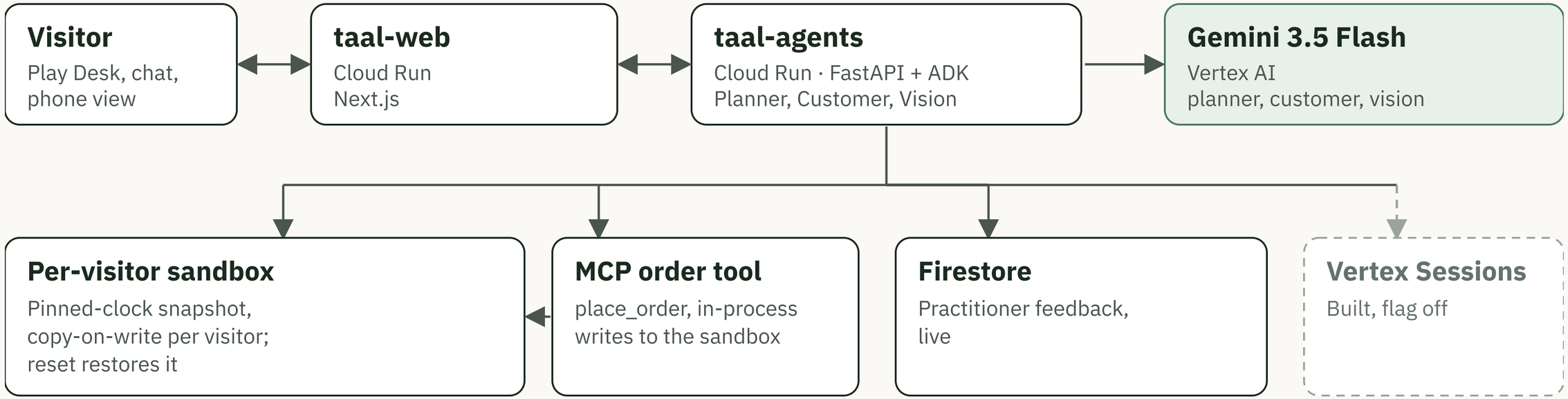



## Chat as Ravi, live app, 16 Oct

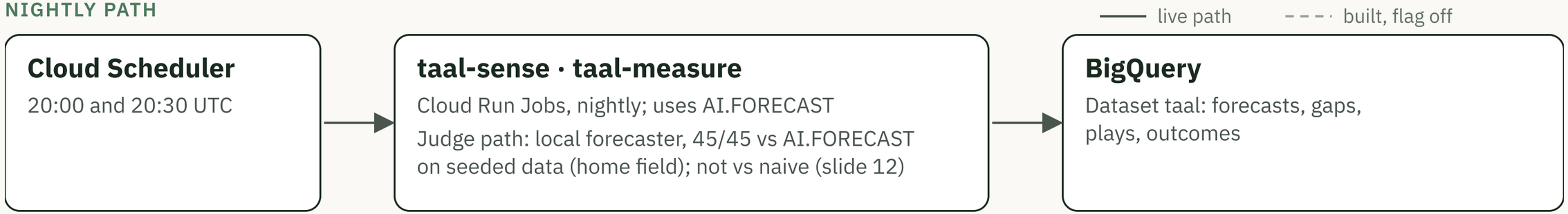
Ravi is in the treated group: the offer arrives, Add to cart places a ₹450 order through the MCP order tool. Replies also come in Kannada with an English gloss.

ARCHITECTURE

# A sandbox serves judges; nightly jobs run on BigQuery.




NIGHTLY PATH



TECHNOLOGIES USED

# Each Google service, and the decision it makes.

| SERVICE | THE DECISION IT MAKES |
|---|---|
| **Google ADK planner (Gemini 3.5 Flash)** | Proposes the play: mechanic, audience, window, in a LoopAgent (max 3) with 6 tools |
| **Gemini on Vertex AI** | Drafts the offer, chats with the customer, reads the pallet photo |
| **Model Armor** | Screens every chat message and reply for injection and jailbreak (live 14 Oct); fails open, code guardrails still apply |
| **BigQuery AI.GENERATE_TABLE** | Pre-generated the offer copy for the 7 seeded plays; stored and reused at Approve. Other plays try one call within 3 s, then templates. |
| **Local forecaster (BigQuery AI.FORECAST nightly)** | Re-forecasts the write-off on Approve; the nightly batch feeds Sense |
| **Firestore** | Stores practitioner feedback, live in the deployed service |
| **Cloud Run, Jobs and Scheduler** | Serves the app and API; runs Sense and Measure nightly; scales to zero when idle |
| **MCP order tool** | Places the customer's order when they accept the offer |
| **Deterministic guardrails and estimator** | Check every play against 8 rules and own every rupee; Gemini never sets a number |

DEPLOY GitHub Actions runs verify, then deploys to Cloud Run on every merge to main.

ESTIMATED IMPLEMENTATION COST

# Priced from measured usage; sold on rupees rescued.

PLANNER COST PER PLAY · 14 OCT

## ₹6.5 / $0.08

**gemini-3.5-flash, token usage priced at list rates. Modelled, not a bill.**

Offer copy is generated once and stored, so Approve on the seeded plays waits on no model call. Services scale to zero between visits. Production path: docs/scale.md.

₹486 / 30 days (23 Sep, pre-switch) was project spend, not per play.

### Forward-looking profiles ESTIMATES

| Profile | Scale | Monthly |
|---|---|---|
| **Micro** | 1 shop, ≤500 SKUs | ₹200–500 |
| **Mid** | 300 SKUs, 10 nodes | ₹2,500–5,000 |
| **Large** | 20,000 SKUs, 500 nodes | ₹1–2 lakh |

BUYER AND PRICING

**Buyer:** the category or supply-chain head, who owns write-off.
**Basis:** a fee per node per month, or a percentage of rupees rescued vs the holdout.

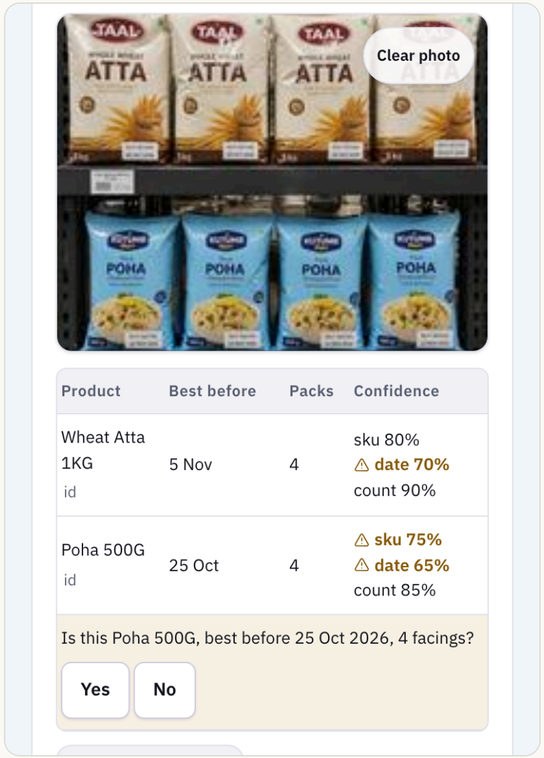
PROJECTED MARGIN LESS RUN COST ILLUSTRATION

**≈ ₹21,710 – ₹156 to ₹313 per node**

Per 28 days at the 5% prior: ₹347,356 margin over 16 nodes, less Mid-profile cost (₹2,500–5,000 a month, estimate). Synthetic.

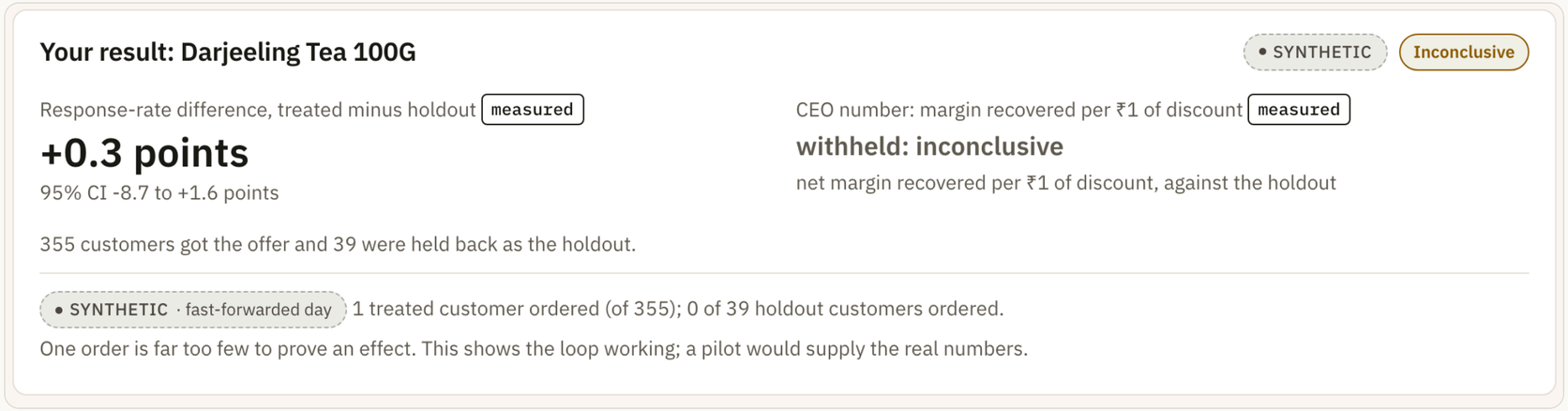
SNAPSHOTS OF THE PROTOTYPE

# Doubtful reads go to a person; outcomes say SYNTHETIC.




## Phone view

A read under 70% confidence is asked back before it reaches inventory.




## Outcomes on a fast-forwarded day, synthetic orders: Tea

355 offered, 39 held out; 1 treated customer ordered, 0 held-out. Response rate +0.3 percentage points, 95% confidence interval –8.7 to +1.6. The range crosses zero, so the page says inconclusive and withholds the rupee figure.

PERFORMANCE REPORT

# Measured and dated, on gemini-3.5-flash.

LIVE PLAN — 11 Oct

**22/24**

plans written by the model on the deployed app, 0 fallbacks; prompt v6, v7 ships

plan_live_2026-10-12.json

CUSTOMER CHAT — 14 Oct

**3.66 s** p50

p95 4.36 s, down from 8.44 s; Model Armor on

customer_latency_2026-10-14_before_after

APPROVE — 15 Oct

**1.4–1.9 s**

warm, recorded play, no model call; ten samples, 13 to 15 Oct; 3.3 s before

approve_timing_2026-10-15.json

PLANNER EVALSETS — 14 Oct

**39/46**

first proposal accepted by the guardrails

live_eval_2026-10-14/summary.json

ADVERSARIAL EVAL — 14 and 16 Oct

**50 of 60** attack runs stopped

0 of 18 genuine requests refused. 10 of 60 runs (4 cases) got through; all 4 closed in code since (V3, P3, V6 re-ran 3/3; C8 refused by the tool).

eval/evaluation.md (adversarial; Hardening, 16 Oct)

PUBLIC BACKTEST, UCI ONLINE RETAIL II — 14 Oct

**WAPE 1.12** vs 0.99 naive seasonal

The forecaster does not beat a naive seasonal baseline here. The loop after the forecast is the product.

eval/external_backtest.md · external_backtest_v2_2026-10-14.json

**Also measured:** photo intake 30/30 dates read on synthetic photos, not real phone photos (14 Oct, vision_synthetic.json); 1,700+ tests (870 pytest + 833 Playwright, collected 16 Oct) in make verify before every deploy. Method and every source: eval/evaluation.md, deck/claims.md.

VALIDATION · PRACTITIONER RESPONSES, SEPTEMBER 2026

# What seven practitioners told us.

**3 of 7**
write off stock weekly or daily

**3 of 6**
online sellers say the sell-by rule limits what they sell

**0 of 7**
measure a promotion against a control group

**4 of 7**
want a one-week pilot

*"Blanket markdowns erode margin on products that may still sell normally."*

Category / merchandising, quick-commerce

*"Item-store recommendations with incremental-margin estimates."*

Category / merchandising, quick-commerce

7 usable responses (10 collected, 3 excluded as flagged in config/feedback_exclusions.json). Anonymous by consent; contacts never published. Counts, not percentages.

USER GUIDE

# Three people, three screens, one approved play.

## Planner

/desk

1. Open the Play Desk and pick a gap.
2. Check the guardrails and counterfactuals.
3. Click Approve. The forecast moves.

## Node manager

/phone

1. Photograph a pallet, or tap a sample.
2. Confirm the rows read below 0.7 confidence.
3. Approve the play from the gap card.

## Customer

/chat

1. Open the chat. It starts with "Any offers today?".
2. Ask about a product. You see a stock band, not a count.
3. Send STOP to withdraw consent.

**Badges.** LIVE: a real call this visit. REPLAY: stored output replayed. SYNTHETIC: fabricated demo data. "Recorded from Gemini": a dated real model run.

IMPACT · WHAT IS MEASURED, WHAT IS PROJECTED

# Projected margin hinges on one assumption.

TWO LOTS · PROJECTED

### Tea (recorded Gemini run)

**₹9,191 recovered**

**Vs doing nothing; projected, seeded tenant. ₹6,186** ahead of a blanket 20% markdown. 103 units, ₹35,020 at stake.

### Chips (10 Oct, API snapshot)

**-₹279**

Projected. 368 units, ₹9,200 at stake. Net −₹8,468 vs −₹8,189.

RESPONSE PRIOR SENSITIVITY · PROJECTED

| PRIOR | MARGIN | PER NODE / YEAR |
|---|---|---|
| 2% | ₹1.41 lakh | ₹1.15 lakh |
| **5% default** | **₹3.47 lakh** | **₹2.83 lakh** |
| 10% | ₹6.19 lakh | ₹5.04 lakh |

Seeded tenant, 28-day run, 550 gaps. Per node / year = margin ÷ 16 nodes × 365/28.

MEASURED VS PROJECTED

**Measured on real customers:** none yet.

**Synthetic holdout, Tea:** 355 treated, 39 held back; 1 responder vs 0, inconclusive.

**Power (SIMULATION):** a 5 pp lift needs about 1,156 customers at a 90/10 split.

**Projected at 5%:** ₹347,356 margin, ₹706,183 waste avoided.

**Practitioners:** 7 usable responses; 3 of 7 write off stock weekly or daily.
**Scale of the problem:** sourced loss figure and arithmetic on slide 2 (as reported, all inventory loss, not a Taal result).

LONGEVITY · RUNNING IT FOR REAL

# What it takes to run beyond the demo.

## Production path

**Today:** services run as the owner-role deploy identity; CI uses a stored key. Secrets sit in a managed store. Fair-use limits cap plan, chat and photo calls per visitor and in total.

**Next:** least-privilege service accounts and workload identity.

Model IDs pinned with retirement dates: gemini-3.5-flash live, Flash-Lite fallback; the previous model retires 20 Oct 2026.

## How we kept it fast

Offer copy generated ahead and stored, so Approve takes 1.4 to 1.9 s warm (recorded play, no model call; 13 to 15 Oct; 3.3 s before).

Chat p95 8.4 s to 4.4 s (14 Oct): stock prefetch, client reuse, minimal thinking.

Services scale to zero; the first call after idle takes longer. Breaks first: max-instances 1 with re-plan state in memory.

## Feasibility

Buyer: category or supply-chain head. Fee per node or rupee share.

Pilot: 50 to 60 testers, one week, not yet run.

Planner: about ₹6.5 per play at list prices (gemini-3.5-flash, 14 Oct); offer copy generated once and stored.

Estimates: ₹200 to 500 (Micro), ₹2,500 to 5,000 (Mid), ₹1 to 2 lakh (Large). ESTIMATE

**CONNECTING TO A RETAILER** The write-off owner (category or dark-store ops) pays. Stock with expiry and sales load from CSVs today; consented customers are a pilot's first step. Opted-in customers only, STOP in code, designed to India's DPDP consent model (not certified). Every play needs human approval.

ADDITIONAL DETAILS · FUTURE DEVELOPMENT

# Next: a real pilot, real photos, more tenants.

### 1 A pre-registered pilot

One week, 50–60 consented testers, 40–50% holdout. Metric pre-registered in docs/pilot.md.

DESIGNED

### 2 Vision on real photos

Harness built. Real camera photos are the missing input; 30/30 (14 Oct) is on synthetic ones.

HARNESS BUILT

### 3 Multi-tenant BigQuery

Tenant id on every table; the BigQuery path is built, flag off until a second tenant.

BUILT, FLAG OFF

LEARNING CURVE · MEASURED, CHANGED, RE-MEASURED

| MEASURED | BEFORE | WHAT WE CHANGED | AFTER |
|---|---|---|---|
| **Chat p95, same 24 prompts** | 8.44 s | stock prefetch, Model Armor client reuse, minimal thinking | **4.36 s, Model Armor on (14 Oct)** |
| **Approve, warm** | 3.3 s (12 Oct) | offer copy generated ahead of Approve | **1.4–1.9 s (13 to 15 Oct, ten samples)** |
| **Live planner** | no_play, 110.9 s (20 Sep, pre-switch) | gap lookup moved into code | **22 of 24 model-authored (11 Oct)** |
| **Planner evalsets** | 43/46 match, 1/46 trajectory (old recordings) | re-recorded on gemini-3.5-flash | **46/46, 18/46: regression guard, not accuracy** |
| **Adversarial eval** | 10 of 60 runs (4 cases) got through | code-only fixes | **all 4 closed in code; V3, P3, V6 re-ran 3/3** |

Rows are copied from docs/learning_curve.md, which also lists the changes that made a number worse. Pre-switch means gemini-2.5-flash.

LINKS

# Taal: the agent that sells what the forecast says you'll throw away

DEMO VIDEO

drive.google.com/file/d/1hBKapntHMnc4FNybRqSEWods2MTrluBw

GITHUB REPOSITORY

https://github.com/Nandish3010/probable-fortnight

LIVE APP

https://taal-web-2obkp776ca-el.a.run.app

Transfer plus targeted nudge: ₹9,191 projected recovery vs doing nothing on one tea lot · seeded tenant

22 of 24 live plans on gemini-3.5-flash, p50 26.5 s (11 Oct)

Holdout so far: synthetic and inconclusive; a real pilot is next

Open the live app, click Run the 60-second beat, then Approve. Nothing you do there is saved.
Then, as Ravi, type: “Is the Darjeeling tea in stock?” · “Do you have Cola Zero?” · “STOP”

**Team Taal** Google Cloud AI Builder Cup 2026 · Retail & Commerce